

PLSQL

◆ What is PLSQL:-

- PL/SQL stands for “Procedural Language extension of SQL” that is used by Oracle.
- PL/SQL is a block structured language that can have multiple blocks in it.
- A line of PL/SQL text contains groups of characters known as lexical units.

➤ Syntax:-

```
DELIMITER //  
BEGIN  
.....  
END //  
DELIMITER ;
```

➤ Key Features of PL/SQL:-

- **Block Structure:** PL/SQL can execute a number of queries in one block using single command.
- **Procedural Constructs:** One can create a PL/SQL unit such as procedures, functions, packages, triggers, and types, which are stored in the database for reuse by applications.
- **Error Handling:** PL/SQL provides a feature to handle the exception which occurs in PL/SQL block known as exception handling block.
- **Reusable Code:** Create stored procedures, functions, triggers, and packages, which can be executed repeatedly.
- **Performance:** Reduces network traffic by executing multiple SQL statements within a single block.

Control Statements:-

◆ if then else:-

- `select check_if_else(10);`
- Example:-

```
DELIMITER //
CREATE FUNCTION check_if_else(val INT)
RETURNS varchar(100)

BEGIN

    DECLARE income_level varchar(100);
    IF val = 10 THEN
        SET income_level= 'value is 10';
    ELSEIF val =20 THEN
        SET income_level = 'value is 20';
    ELSE
        SET income_level = 'Incorrect value';
    END IF;
    RETURN income_level;

END; //
DELIMITER ;
```

Control Statements:-

◆ Case :-

➤ **select** check_case(1000);

➤ **Example:-**

```
DELIMITER //  
CREATE FUNCTION check_case(val int)  
RETURNS varchar(100)  
  
BEGIN  
  
    DECLARE income_level varchar(100);  
    CASE val  
        WHEN 1000 THEN  
            SET income_level = 'Low Income';  
        WHEN 5000 THEN  
            SET income_level = 'Avg Income';  
        ELSE  
            SET income_level = 'High Income';  
    END CASE;  
    RETURN income_level;  
  
END; //  
DELIMITER ;
```

FUNCTION

◆ What is Function:-

- A function is same as a procedure except that it returns a value.
- A function must always return a value, and on the other hand a procedure may or may not return a value.
- **Syntax:-**

DELIMITER //

CREATE FUNCTION function_name(parameters....)
RETURNS datatype

BEGIN
 statements
 Return value;
END //

DELIMITER ;

➤ **Example:-**

- **select** my_square()

DELIMITER //

CREATE FUNCTION my_square(val int)
RETURNS int

BEGIN
 DECLARE result int;
 Set result=0;
 Set result=val * val;
 Return result;
END //

DELIMITER ;